

Problem Set 7: Techniques of Integration

Math Mb — Integral Calculus

Spring 2026

Review of Key Formulas

Recall the following integration formulas:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \quad \int e^x dx = e^x + C, \quad \int \frac{1}{x} dx = \ln|x| + C. \quad (1)$$

$$\int \sin x dx = -\cos x + C, \quad \int \cos x dx = \sin x + C, \quad \int \sec^2 x dx = \tan x + C. \quad (2)$$

Integration by Parts:

$$\int u dv = uv - \int v du.$$

Part I: u -Substitution

1. Evaluate each integral using substitution.

(a) $\int 3x^2 \cos(x^3) dx$

(b) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$

(c) $\int_0^1 \frac{x}{(1+x^2)^2} dx$

(d) $\int \frac{\ln x}{x} dx$

Part II: Integration by Parts

2. Evaluate each integral using integration by parts.

(a) $\int x e^{2x} dx$

(b) $\int x^2 \sin x \, dx$

(c) $\int \ln x \, dx$

(d) $\int e^x \cos x \, dx$

3. **Worked Example.** Here is a detailed solution for $\int x^2 e^x \, dx$, applying integration by parts twice:

$$\begin{aligned} \int x^2 e^x \, dx &= \underbrace{x^2}_u \cdot \underbrace{e^x}_v - \int \underbrace{e^x}_v \cdot \underbrace{2x \, dx}_{du} \\ &= x^2 e^x - 2 \int x e^x \, dx \\ &= x^2 e^x - 2 \left[x e^x - \int e^x \, dx \right] \\ &= x^2 e^x - 2x e^x + 2e^x + C \\ &= e^x (x^2 - 2x + 2) + C. \end{aligned}$$

Part III: Partial Fractions

4. Evaluate each integral using partial fraction decomposition.

(a) $\int \frac{5x + 3}{(x + 1)(x - 2)} \, dx$

(b) $\int \frac{3x^2 + 2}{x(x^2 + 1)} \, dx$

(c) $\int \frac{x^3 + 2}{x^2 - 1} \, dx$ (*Hint: Perform polynomial long division first.*)

5. Show the partial fraction decomposition (without evaluating the integral):

$$\frac{2x^2 + x + 4}{(x + 1)(x^2 + 4)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + 4}.$$

Find A , B , and C .

Part IV: Trigonometric Integrals

6. Evaluate:

(a) $\int \sin^3 x \cos^2 x \, dx$

(b) $\int \tan^4 x \sec^2 x \, dx$

(c) $\int_0^{\pi/2} \cos^2 \theta \, d\theta$

Part V: Improper Integrals

7. Determine whether each improper integral converges or diverges. If it converges, find its value.

(a) $\int_1^{\infty} \frac{1}{x^3} \, dx$

(b) $\int_0^{\infty} e^{-2x} \, dx$

(c) $\int_0^1 \frac{1}{\sqrt{x}} \, dx$

(d) $\int_{-\infty}^{\infty} \frac{1}{1+x^2} \, dx$